Enhancing a Real-Time File-Sharing Application Through Advanced Software Engineering Practices

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Executive Summary

The report offers redesigning and upgrading of an already existing open-source file-sharing platform initially forked on GitHub. The hereditary one showed poor architectural underpinning, failed back-end communication, and missing real-time capability. An organized engineering methodology was implemented to enhance the logic of the back-end, front-end communication, scalability, and stability as well as real-time file streaming with the help of WebSockets.

Significant improvements were the complete creation of a lobby, peer communication in rooms, secure and consistent temporary file storage, the enhanced modular architecture, and synchronization of a React interface with a Node.js backend. It also incorporated the use of a version control tracking using Git, extensive testing processes, software metrics optimization, and the optimization of software, as well as a COCOMO-based cost estimation methodology. The result indicates the adherence to high standards of software engineering, such as modular design, constant refinement, structured testing, and performance-oriented refactoring. Additional suggestions are cloud implementation, database persistence, and scalability in big files.

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1. Introduction

Real-time file-sharing software is used to enable quick transfer of data between and among users of the network and is currently sought after in collaborative setups, such as in education, remote work, and distributed development teams. The initial open-source app that was chosen to be improved was conceptually based, containing Socket.io events, a React user interface, and server code, but most of the features were not complete, unstable, or unconnected.

The redevelopment was done in a systematic advanced software engineering method, which tackled software quality concerns with the refinement of requirements, redesign of architecture, modular implementation, and test-driven performance. The prioritization items included creating stable communication between a client and a server, which provided the possibility of real-time interaction of many users, enhancing the behavior of file management, and removing old and unnecessary technical debt.

The report includes a critical analysis of limitations of the inherited codebase, the reasons behind refactoring decisions, development procedures, testing and metrics, and an evaluation of the improvement of software quality according to the engineering best practices.

2. Evaluation of the Existing Forked Repository

2.1 Lack of Functional Backend Integration

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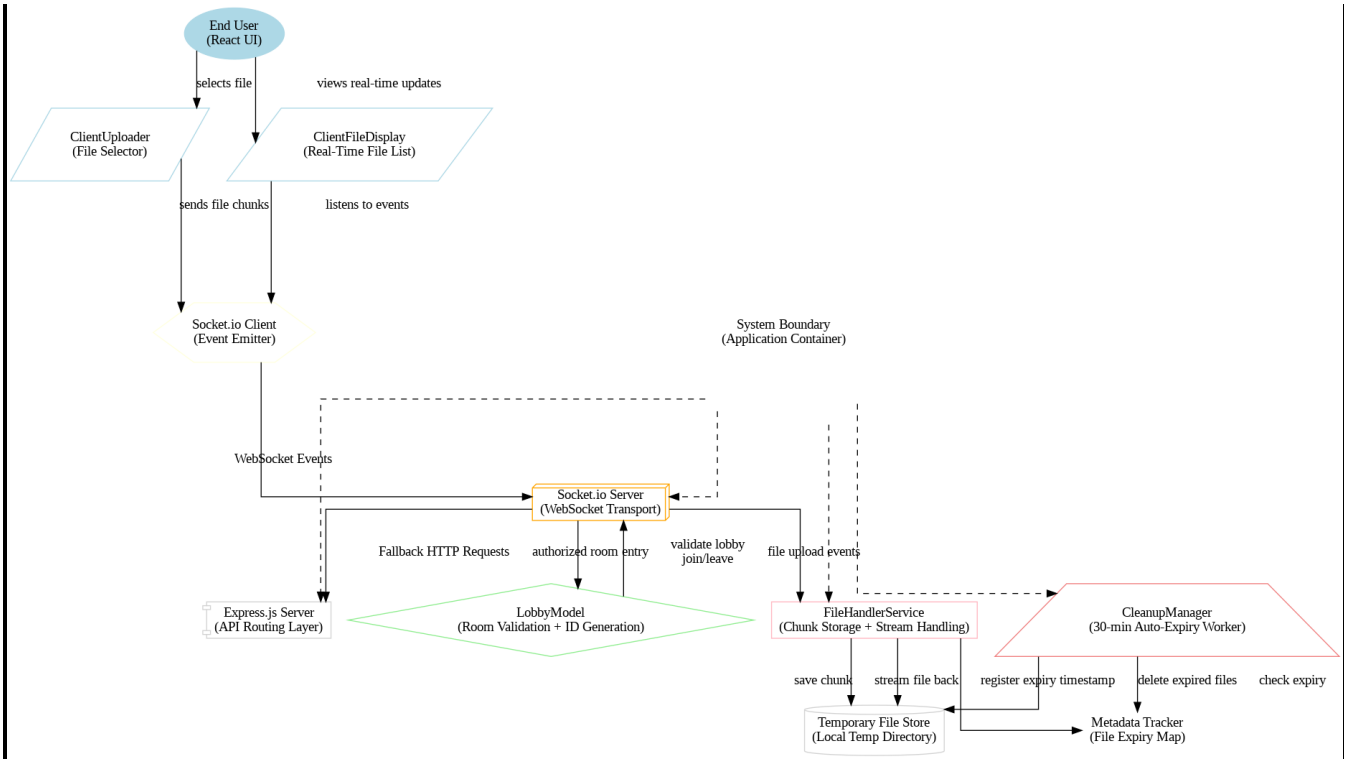

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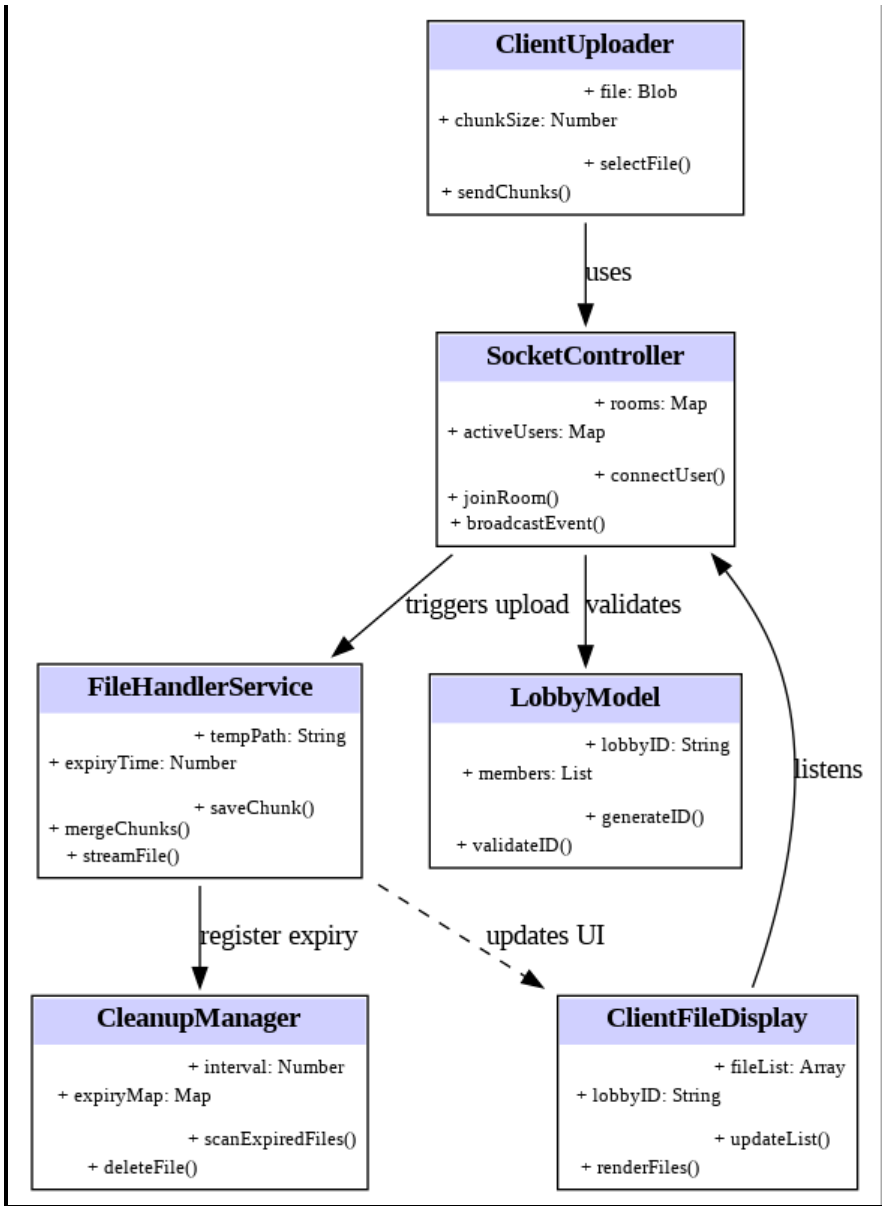

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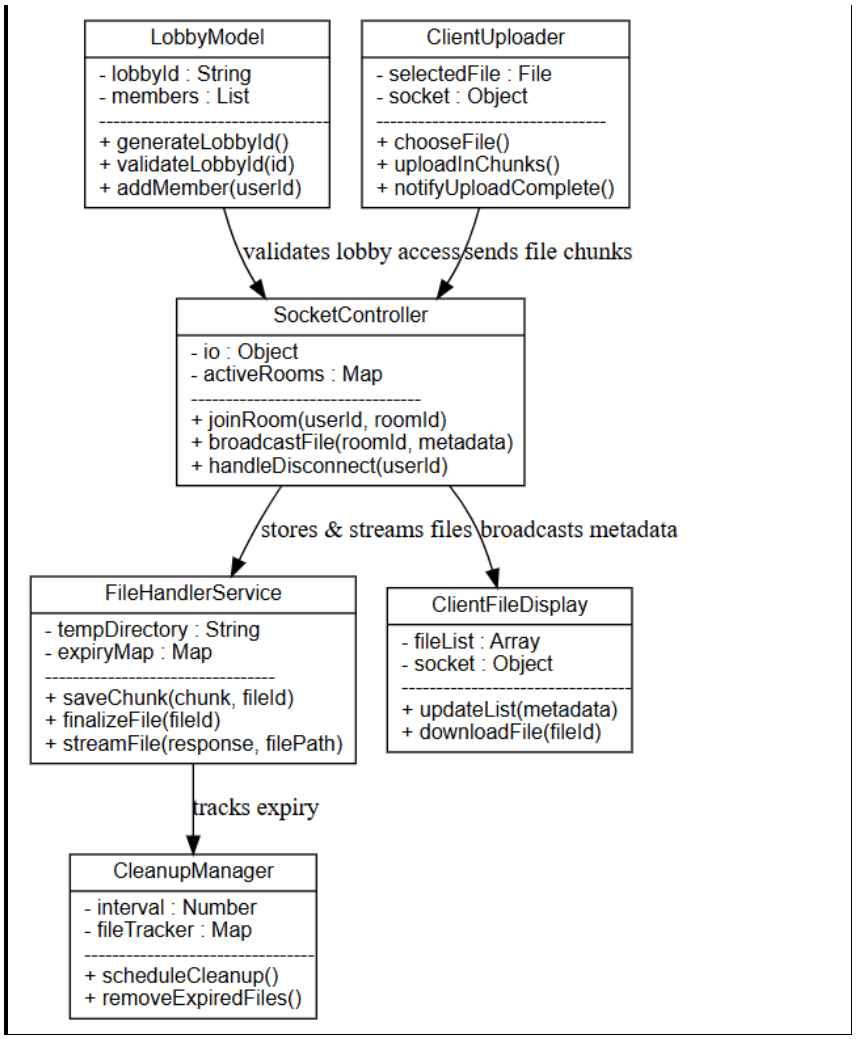

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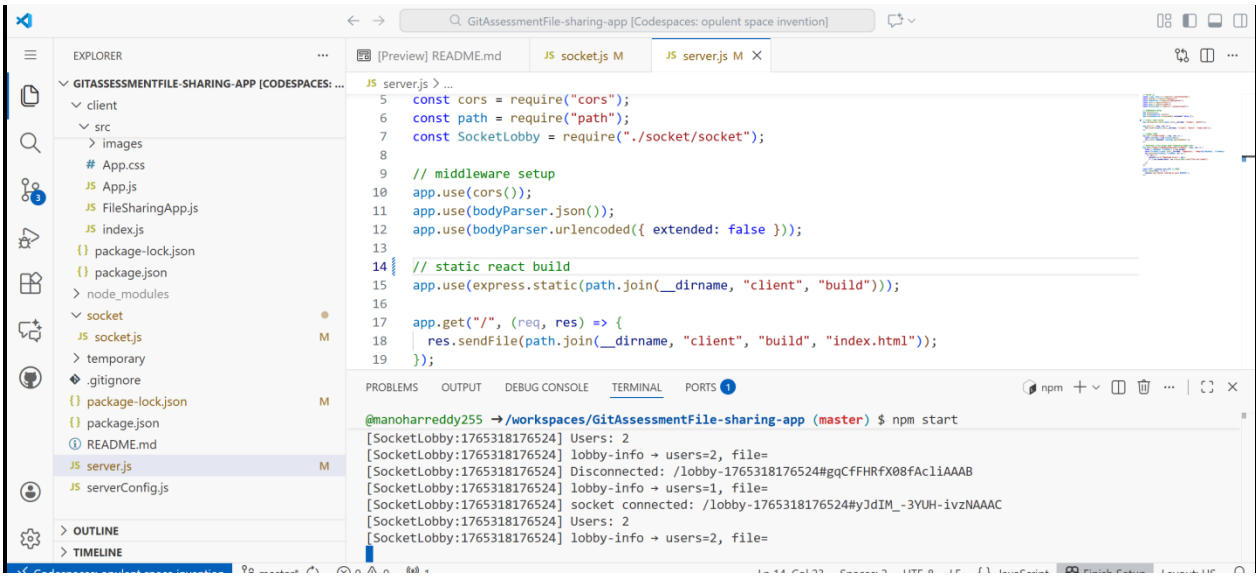
socket > JS socket.js > ...

```
1  const { http } = require("../serverConfig");
2  const io = require("socket.io")(http, {
3    maxHttpBufferSize: 100 * 1024 * 1024
4  });
5  const ss = require("socket.io-stream");
6  const path = require("path");
7  const fs = require("fs");
8  const util = require("util");
9
10 const unlink = util.promisify(fs.unlink);
11 const readdir = util.promisify(fs.readdir);
12
13 // store namespaces to avoid duplicates
14 const socketLobbies = new Map();
15
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 1

Port	Forwarded Address	Running Process	Visibility	Origin
8000	https://o...	node server.js (6732)	Private	Auto Forwarded

Add Port



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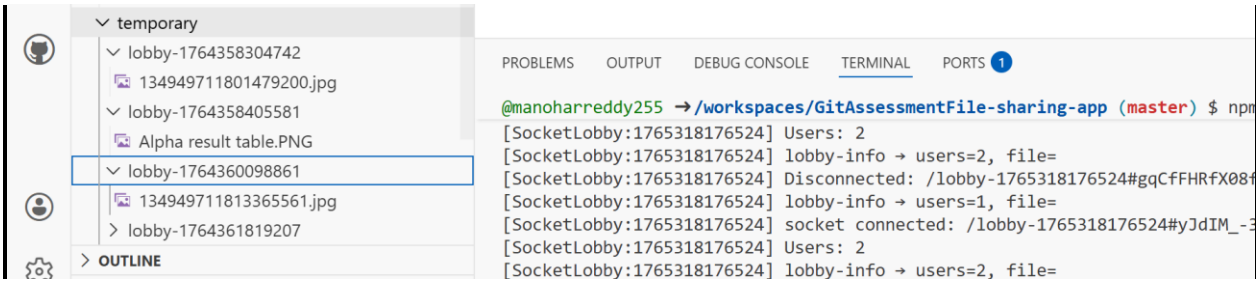

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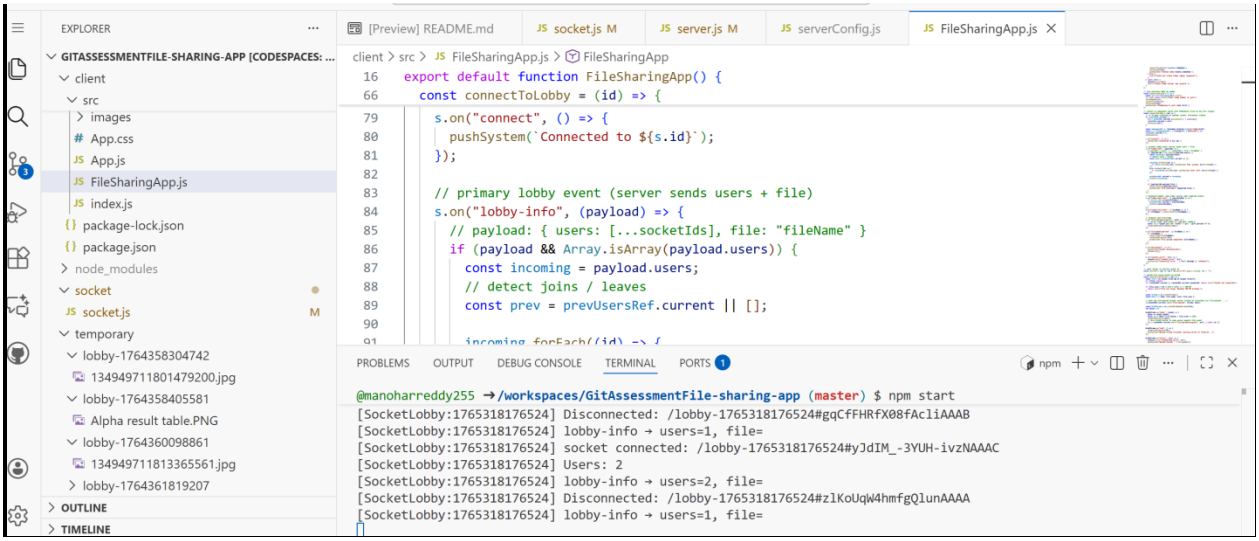

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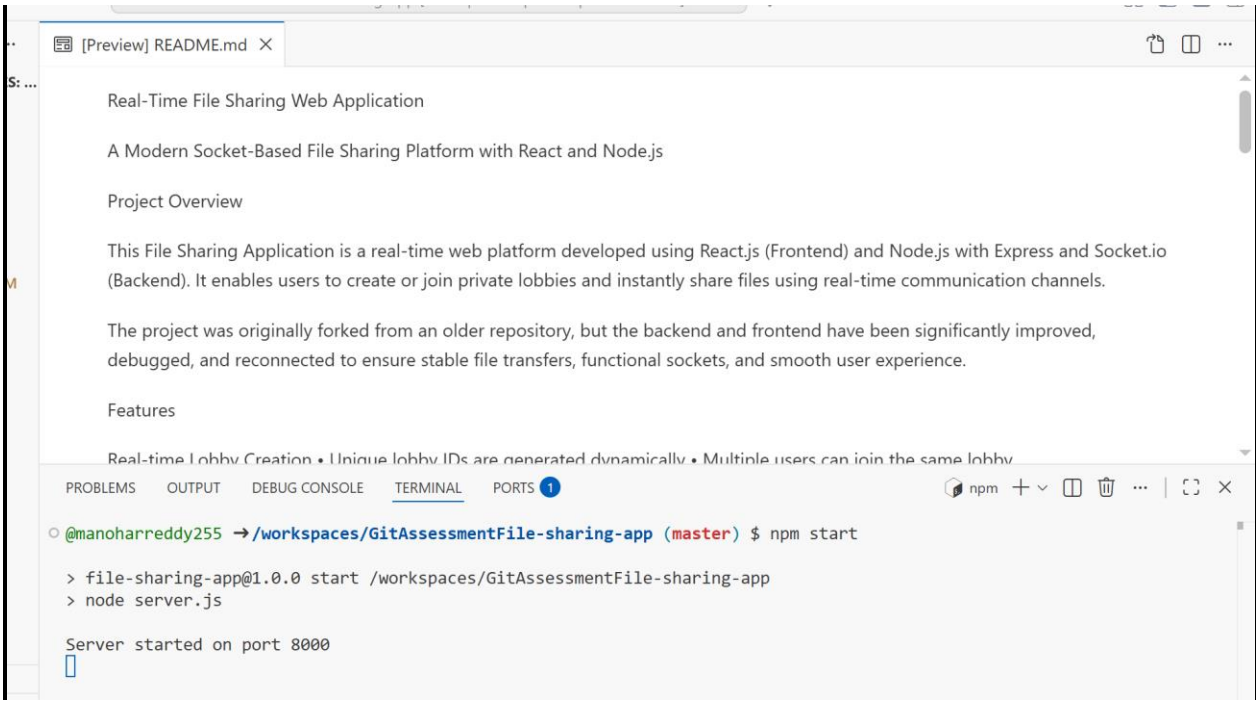

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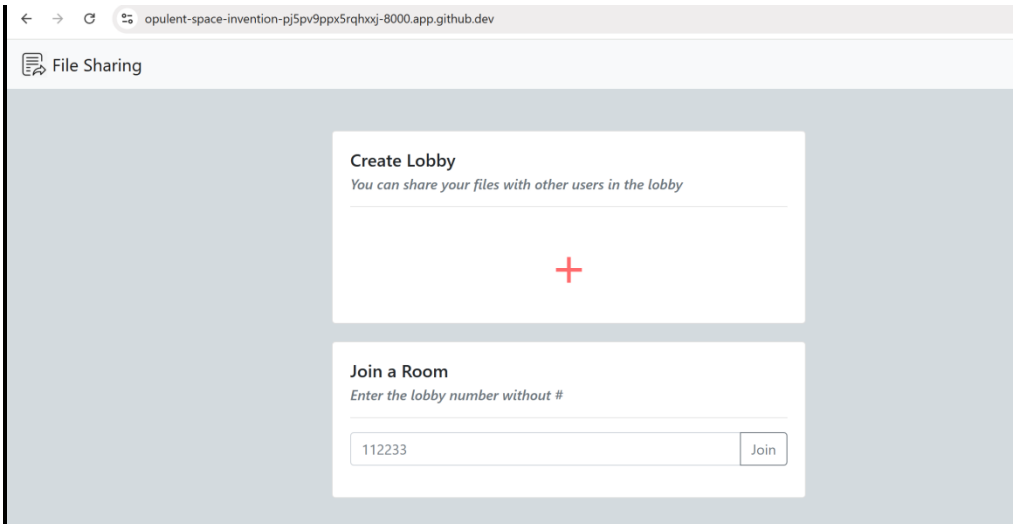

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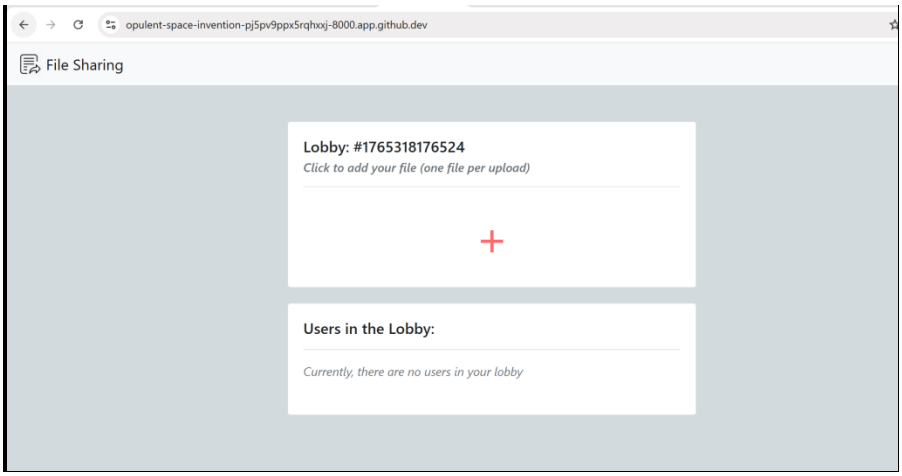

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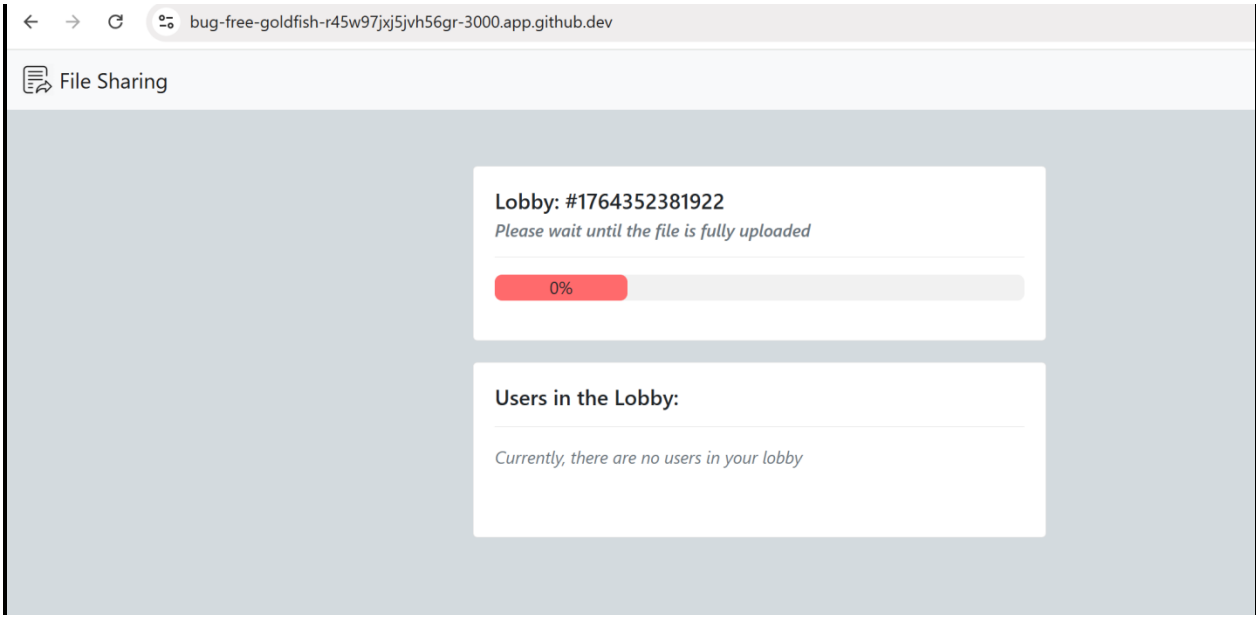

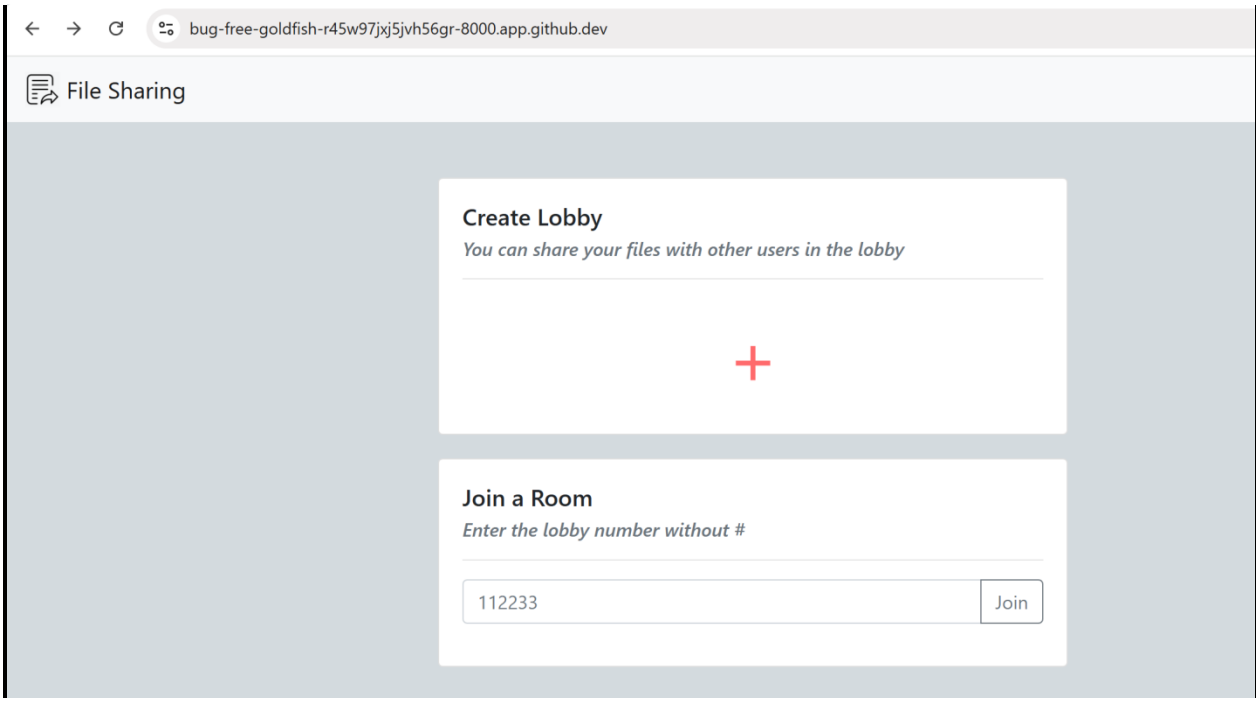
bug-free-goldfish-r45w97jxj5jvh56gr-8000.app.github.dev
File Sharing
Create Lobby
You can share your files with other users in the lobby
Join a Room
Enter the lobby number without #
112233
Join

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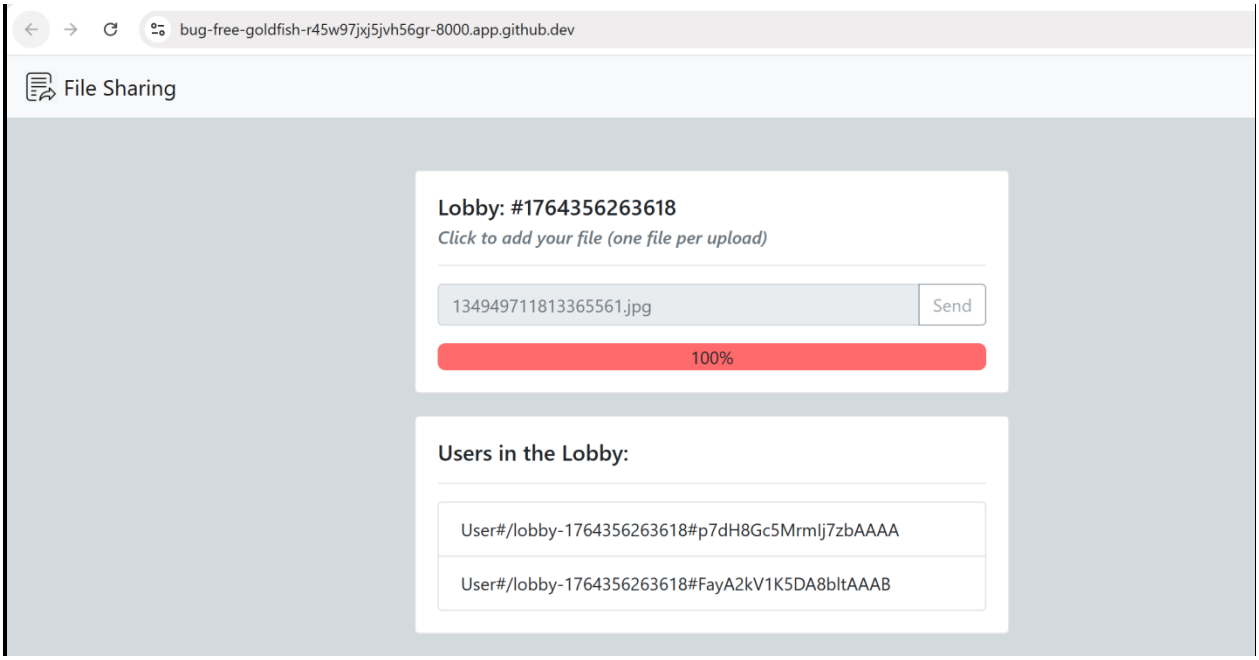
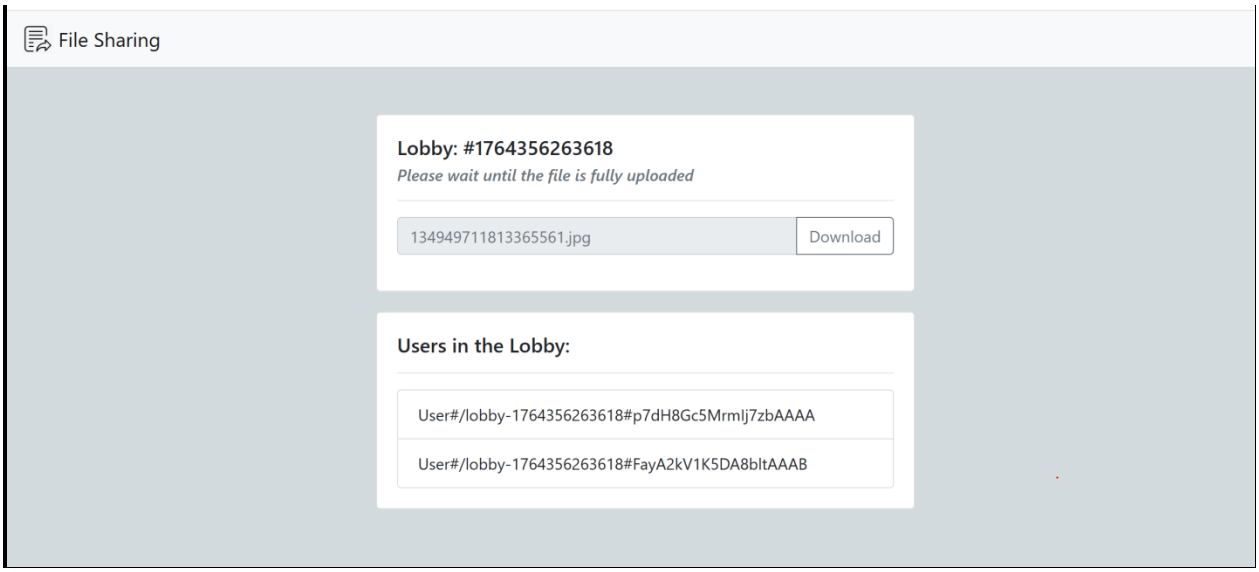
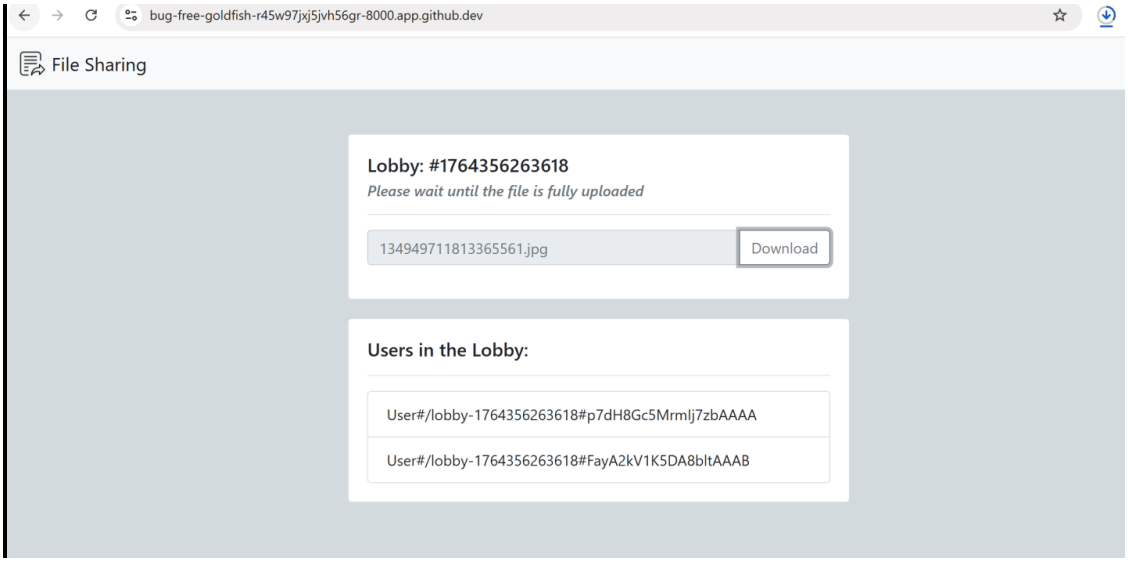

bug-free-goldfish-r45w97jxj5jvh56gr-8000.app.github.dev
File Sharing
Lobby: #1764356263618
Click to add your file (one file per upload)
134949711813365561.jpg
Send
100%
Users in the Lobby:
User#/lobby-1764356263618#p7dH8Gc5MrmIj7zbAAAA
User#/lobby-1764356263618#FayA2kV1K5DA8bltAAAB


File Sharing
Lobby: #1764356263618
Please wait until the file is fully uploaded
134949711813365561.jpg
Download
Users in the Lobby:
User#/lobby-1764356263618#p7dH8Gc5MrmIj7zbAAAA
User#/lobby-1764356263618#FayA2kV1K5DA8bltAAAB


bug-free-goldfish-r45w97jxj5jvh56gr-8000.app.github.dev
File Sharing
Lobby: #1764356263618
Please wait until the file is fully uploaded
134949711813365561.jpg
Download
Users in the Lobby:
User#/lobby-1764356263618#p7dH8Gc5MrmIj7zbAAAA
User#/lobby-1764356263618#FayA2kV1K5DA8bltAAAB

```
@user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $ git status
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   client/package-lock.json
        modified:   client/package.json
        modified:   client/src/App.js
        modified:   client/src/Components/CreateLobbyCard/CreateLobbyCard.js
        modified:   client/src/Components/DownloadFileCard/DownloadFileCard.js
        modified:   client/src/Components/FileSharingBox/FileSharingBox.js
        modified:   client/src/Components/JoinToLobbyCard/JoinToLobbyCard.js
        modified:   client/src/Components/UploadFileCard/UploadFileCard.js
        modified:   client/src/Components/UserListCard/UserListCard.js
        modified:   package-lock.json
        modified:   package.json
        modified:   server.js
        modified:   serverConfig.js
        modified:   socket/socket.js

Untracked files:
  (use "git add <file>..." to include in what will be committed)
```

```
● @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $ git add .
● @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $ git commit -m "Fixed backend route
s and socket namespace issues"
[master 7fba9ed] Fixed backend routes and socket namespace issues
 20 files changed, 15234 insertions(+), 10726 deletions(-)
 create mode 100644 client/src/Components/ChatArea/ChatArea.js
 create mode 100644 client/src/FileSharingApp.js
 create mode 100644 temporary/lobby-1764358304742/134949711801479200.jpg
 create mode 100644 temporary/lobby-1764358405581/Alpha result table.PNG
 create mode 100644 temporary/lobby-1764360098861/134949711813365561.jpg
 create mode 100644 temporary/lobby-1764361819207/134949711813365561.jpg
○ @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $
0 ⚠ 0  0   Ln 155, Col 1   Spaces: 2   UTF-8   LF   { } JavaScript   Finish Setup
```

```
● @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $ git add .
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 create mode 100644 temporary/lobby-1764358304742/134949711801479200.jpg
 create mode 100644 temporary/lobby-1764358405581/Alpha result table.PNG
 create mode 100644 temporary/lobby-1764360098861/134949711813365561.jpg
 create mode 100644 temporary/lobby-1764361819207/134949711813365561.jpg
● @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $ git push origin master
Enumerating objects: 59, done.
Counting objects: 100% (59/59), done.
Delta compression using up to 2 threads
Compressing objects: 100% (28/28), done.
Writing objects: 100% (36/36), 4.53 MiB | 10.29 MiB/s, done.
Total 36 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 4 local objects.
To https://github.com/user2024-png/GitAssessmentFile-sharing-app
   b49ea1c..7fba9ed  master -> master
○ @user2024-png ➜ /workspaces/GitAssessmentFile-sharing-app (master) $
0                                   Ln 155, Col 1   Spaces: 2   UTF-8   LF   { } JavaScript   Finish Setup   L
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